

# Searching Techniques: Linear Search & Binary Search Data Structures

> **Definition:** Searching is the algorithmic process of locating the position of a target key element `K` within a collection of items (such as an array). The two primary searching techniques are **Linear Search** and **Binary Search**.

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## 1. Detailed Technical Explanation

### 1. Linear Search (Sequential Search)

- **Concept:** Sequentially inspects every element in the array from index `0` to `N - 1` until the target element is found or the end of array is reached.

- **Runtime Complexity:**  $O(N)$  (Linear time complexity)

```
```c
// Linear Search Implementation
int linearSearch(int arr[], int n, int key) {
    for (int i = 0; i < n; i++) {
        if (arr[i] == key) {
            return i;
        }
    }
    return -1;
}
```

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### 2. Binary Search (Divide and Conquer)

- **Concept:** Repeatedly divides the search interval in half by comparing the target key with the middle element `arr[mid]`:

- **Runtime Complexity:**  $O(\log N)$  (Logarithmic time complexity)

```
```c
// Binary Search Implementation
int binarySearch(int arr[], int n, int key) {
    int low = 0, high = n - 1;
    while (low <= high) {
        int mid = (low + high) / 2;
        if (arr[mid] == key) {
            return mid;
        }
        else if (arr[mid] < key) {
            low = mid + 1;
        }
        else {
            high = mid - 1;
        }
    }
    return -1;
}
```

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2. Comparison: Linear Search vs Binary Search

| Feature | Linear Search | Binary Search |

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3. Quick Recall Flow

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Linear